# Supplementary Material

Coarse Observability Across High-Order Finite-Volume Reconstructions: Universal Rank, Conditioning, and Dissipative Memory

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This Supplementary Material collects two blocks of secondary detail relocated from the main article to keep the article self-contained while shortening it: the exact coefficient polynomials of the composed WENO5-LIN + SSP-RK3 one-step operator, and the full one-at-a-time sensitivity analysis behind the nonlinear-rank robustness statement. Both are machine-generated by `make_tables.py` from the same exact rational-arithmetic representations and CSV artifacts as the in-article tables; section and equation numbers of the form "6", "Table 6", "Appendix A" below refer to the main article.

## S1 The SSP-RK3 composition

For a linear autonomous update the Shu–Osher stages reduce to the classical third-order stability polynomial: with $A_{\mathrm{FE}} = I + Z$ and $Z = A_{\mathrm{FE}} - I$ the ($\lambda$-scaled) flux-difference part of Appendix A.2 of the main article,

$$A_{\mathrm{RK3}} = I + Z + \tfrac{1}{2}\,Z^2 + \tfrac{1}{6}\,Z^3,$$

a circulant with support $s \in \{-6, \dots, 9\}$, i.e. $(m_L, m_R) = (9, 6)$ and total reach $w = 15$. Table S1 lists the sixteen cubic coefficient polynomials. The outermost coefficients $c_9 = \lambda^3/162000$ and $c_{-6} = \lambda^3/48000$ are nonzero for every $\lambda > 0$, so the reaches and collar counts of the pairing are $\lambda$-independent on $(0, 1]$ (cf. Remark 1 of the main article).

Table S1: Exact coefficient polynomials of the composed WENO5-LIN + SSP-RK3 one-step operator (sixteen cells, paired columns).

| $s$ | $c_s(\lambda)$ | $s$ | $c_s(\lambda)$ |
|---|---|---|---|
| 9 | $\frac{1}{162000}\lambda^3$ | 1 | $\lambda - \frac{31}{150}\lambda^2 - \frac{49}{200}\lambda^3$ |
| 8 | $-\frac{1}{7200}\lambda^3$ | 0 | $1 - \frac{1}{3}\lambda - \frac{329}{720}\lambda^2 + \frac{10211}{64800}\lambda^3$ |
| 7 | $\frac{23}{14400}\lambda^3$ | −1 | $-\frac{1}{2}\lambda + \frac{13}{60}\lambda^2 + \frac{6253}{72000}\lambda^3$ |
| 6 | $\frac{1}{1800}\lambda^2 - \frac{961}{86400}\lambda^3$ | −2 | $\frac{1}{20}\lambda + \frac{13}{120}\lambda^2 - \frac{1849}{28800}\lambda^3$ |
| 5 | $-\frac{1}{120}\lambda^2 + \frac{121}{2400}\lambda^3$ | −3 | $-\frac{1}{40}\lambda^2 - \frac{9}{800}\lambda^3$ |
| 4 | $\frac{31}{480}\lambda^2 - \frac{427}{3000}\lambda^3$ | −4 | $\frac{1}{800}\lambda^2 + \frac{7}{1200}\lambda^3$ |
| 3 | $\frac{1}{30}\lambda - \frac{47}{180}\lambda^2 + \frac{9467}{43200}\lambda^3$ | −5 | $-\frac{1}{1600}\lambda^3$ |
| 2 | $-\frac{1}{4}\lambda + \frac{17}{30}\lambda^2 - \frac{449}{9600}\lambda^3$ | −6 | $\frac{1}{48000}\lambda^3$ |

## S2 Sensitivity of the numerical ranks

The nonlinear ranks of Section 7.6 of the main article depend on declared numerical choices: the rank floor $\theta$, the ensemble size $M$, the random directions, the finite-difference step $\delta$ of the

linearized ladder, and the physical amplitude $\varepsilon$. Experiment E6s varies each knob one at a time around the E6 baseline ($\theta = 10^{-3}$, $M = 400$ plus the deep pair, seed 0, $\delta = 10^{-5}$, $\varepsilon = 10^{-5}$) over all 21 (scheme, regime) cells and compares, per axis setting, the leading-floor ladder $\rho_\theta^{\mathrm{emp}}(L)$ for $L \le 5$ and the horizon-12 rank against the baseline; for the FD-step axis the compared quantity is the linearized ladder $\rho^{\mathrm{lin}}$. Table S2 counts the cells that change.

Table S2: One-at-a-time sensitivity of the E6 ranks (E6s): number of the 21 (scheme, regime) cells whose leading-floor ladder $\rho_\theta^{\mathrm{emp}}(L)$, $L \le 5$, or horizon-12 rank differs from the axis baseline ($\theta = 10^{-3}$, $M = 400$, seed 0, $\delta = 10^{-5}$, $\varepsilon = 10^{-5}$), and the largest change in the horizon-12 rank. The FD-step rows compare the linearized ladders $\rho^{\mathrm{lin}}$ (tolerance $10^{-8}$) instead. The $M$ axis uses nested direction prefixes of one bank; the seed axis redraws all $M = 400$ directions. Body machine-generated by `make_tables.py` from `nonlinear_sensitivity.csv`.

| | | cells changed (of 21) | | |
|---|---|---|---|---|
| axis | setting | ladder $L \le 5$ | $\rho(12)$ | max $\lvert\Delta\rho(12)\rvert$ |
| rank floor $\theta$ | $10^{-2}$ | 11 | 11 | 7 |
| | $10^{-4}$ | 1 | 12 | 4 |
| ensemble size $M$ | 100 | 0 | 1 | 1 |
| | 200 | 0 | 1 | 1 |
| | 800 | 0 | 0 | 0 |
| seed | 1 | 0 | 1 | 1 |
| | 2 | 0 | 0 | 0 |
| FD step $\delta$ (linearized) | $10^{-4}$ | 0 | 0 | 0 |
| | $10^{-6}$ | 0 | 0 | 0 |
| amplitude $\varepsilon$ | $10^{-2}$ | 13 | 17 | 14 |
| | $10^{-3}$ | 4 | 14 | 6 |
| | $10^{-7}$ | 0 | 1 | 2 |

The methodological knobs are inert. Growing the ensemble from 100 to 800 members, redrawing all directions (seeds 1, 2), and moving the FD step across two orders of magnitude leave every reported ladder unchanged; the only motion anywhere is a single borderline cell (MUSCL superbee, smooth regime) whose horizon-12 rank flips between 6 and 7 at $M \le 200$ and seed 1 — one direction sitting at the declared floor. In particular, the plateau staircase of Table 6 persists identically across every ensemble size and seed. The rank floor behaves as a declared tolerance should: tightening it to $10^{-4}$ touches one ladder cell (the same superbee cell, $6 \to 7$ at $L = 5$) but raises the horizon-12 rank in 12 cells (by up to 4) as near-floor directions get counted, and loosening it to $10^{-2}$ drops weakly exposed directions in 11 cells — which is why every rank in Section 7.6 of the main article is quoted at a stated floor rather than as an absolute.

The amplitude axis moves ranks only where the physics in Section 7.6 requires it. At $\varepsilon = 10^{-7}$, the ladders are identical to the baseline (one WENO-Z horizon-12 cell loses two near-floor directions). At $\varepsilon = 10^{-3}$ — the predicted crossover $\varepsilon \approx \sqrt{\epsilon_w}$ for $\epsilon_w = 10^{-6}$ — exactly the WENO5-JS ladders (extremum and jump regimes) and the WENO5-M jump ladder leave the linear-budget shape, while WENO-Z, whose crossover lies far below, keeps its ladders. At $\varepsilon = 10^{-2}$, the smooth- and extremum-regime cells respond broadly, as in Table 6. Throughout, the four MUSCL jump-regime (flat-plateau) ladders are unchanged at *every* amplitude from $10^{-7}$ to $10^{-2}$: the scale-freeness of the positively homogeneous plateau response (Section 6.2 of the main article) is exact in the sweep.